PROJECT PART B

Project 6b: JFET Output Characteristics Analysis

Simulation, Observation, and Technical Analysis using LTspice

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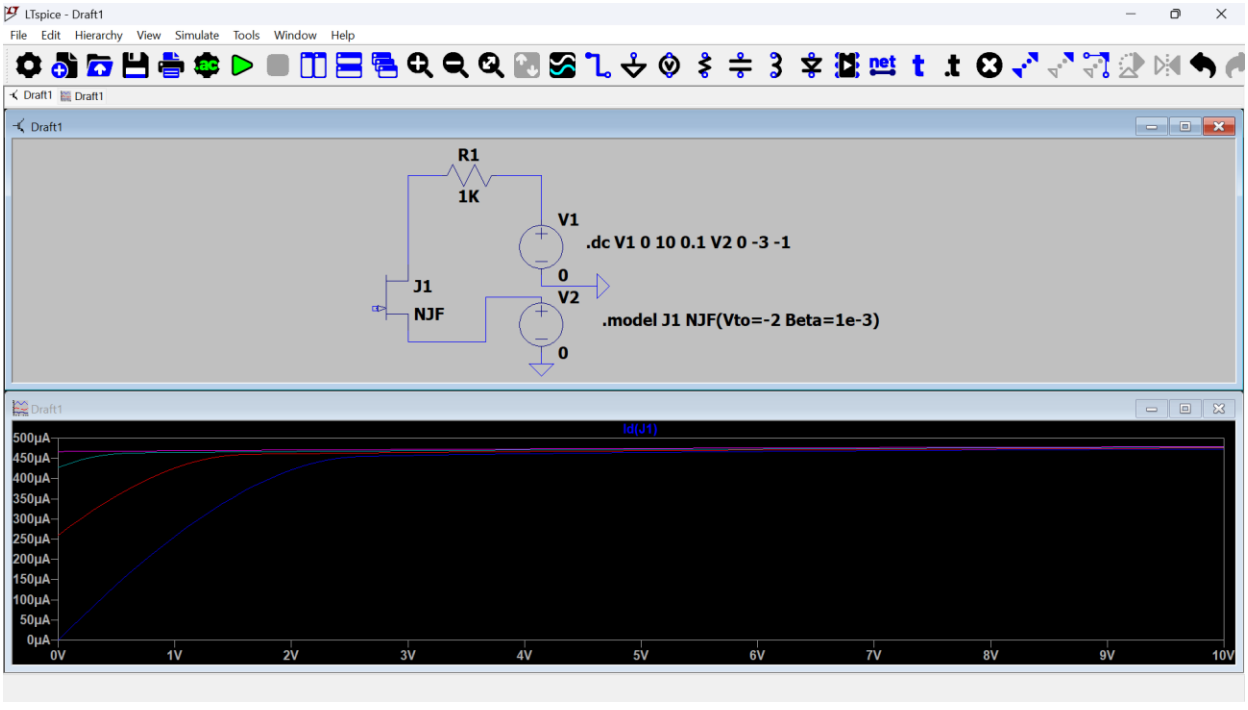
1. Objective & Simulation Setup

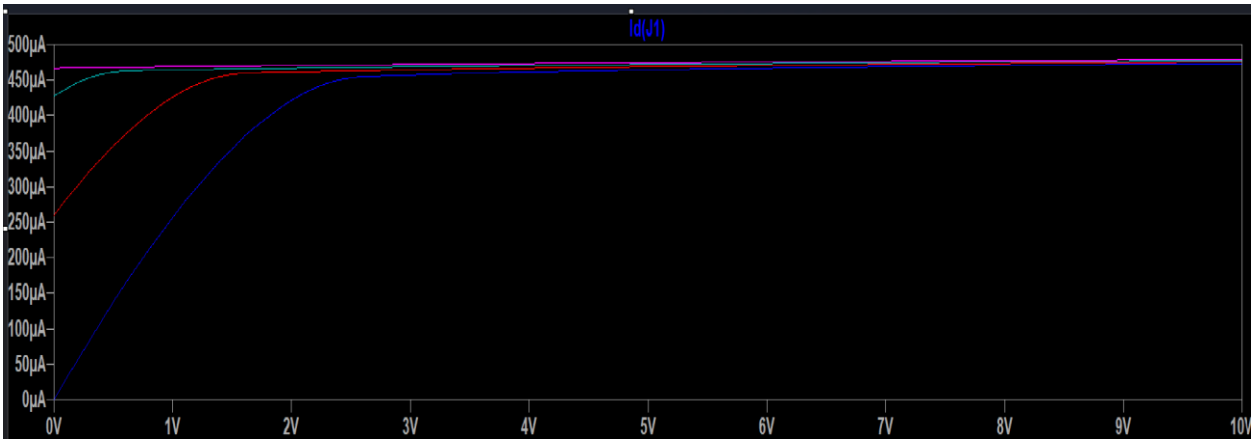
The objective of this task is to simulate and observe the output characteristics of an N-Channel JFET in LTspice, analyze its region of operation, and verify its key electronic behaviors.

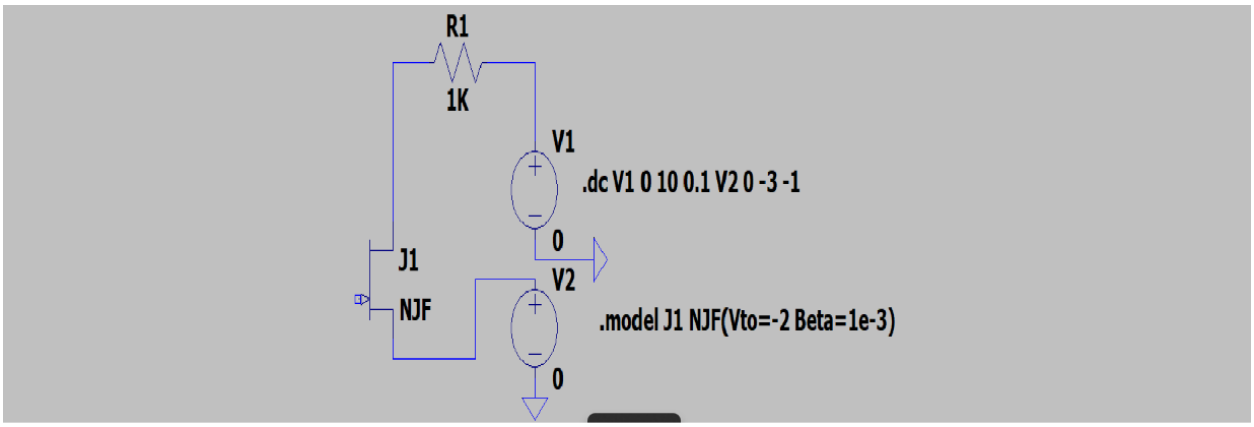
Parameter	Simulation Value / Command
Drain Voltage Source (V1 Sweep)	0V to 10V (Increment: 0.1V)
Gate Voltage Source (V2 Step)	0V to -3V (Step: -1V)
SPICE Directive Command	.dc V1 0 10 0.1 V2 0 -3 -1

2. Simulation Schematic & Output Waveform

Below are the captured LTspice circuit schematic and the resulting output characteristic curves (I_d vs V_ds):







3. Technical Explanations & Inferences

(i) JFET as a Voltage-Controlled Current Source

Explanation: In the saturation (pinch-off) region of the output characteristics curve, the drain current (Id) becomes horizontal and remains practically constant, independent of further increases in drain-to-source voltage (V_ds). Instead, the magnitude of Id is determined entirely by the gate-to-source voltage (V_gs). By varying V_gs, the width of the depletion region inside the channel changes, thereby controlling the flow of charge carriers. Because an input voltage (V_gs) directly controls the output current (Id), the JFET acts as a Voltage Controlled Current Source (VCCS).

(ii) High Input Impedance of JFET

Explanation: Under normal operating conditions, the gate-source p-n junction of an N-channel JFET is reverse-biased ($V_{gs} \leq 0V$). Because a reverse-biased p-n junction allows only an extremely tiny reverse leakage current (in the range of nano-amperes or pico-amperes) to flow through the gate terminal, the input gate current (I_g) is virtually zero. By Ohm's Law, the input impedance ($Z_{in} = \Delta V_{gs} / \Delta I_g$) is extremely high typically ranging from tens of Mega-ohms (MΩ) to Giga-ohms (GΩ). This high input impedance ensures that the JFET draws negligible loading current from preceding circuit stages.

4. Conclusion

The LTspice DC sweep simulation successfully verified the operational behavior of the JFET. The output characteristic curves clearly illustrate the Ohmic region at low V_ds voltages and the Saturation region at higher V_ds voltages, confirming that the JFET functions as a voltage-controlled current source with high input impedance.